Every force has a source and a receiver

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Receiver / ES: objeto que recibe la fuerza / PT: objeto que recebe a força

The object a force acts on; also name its source.

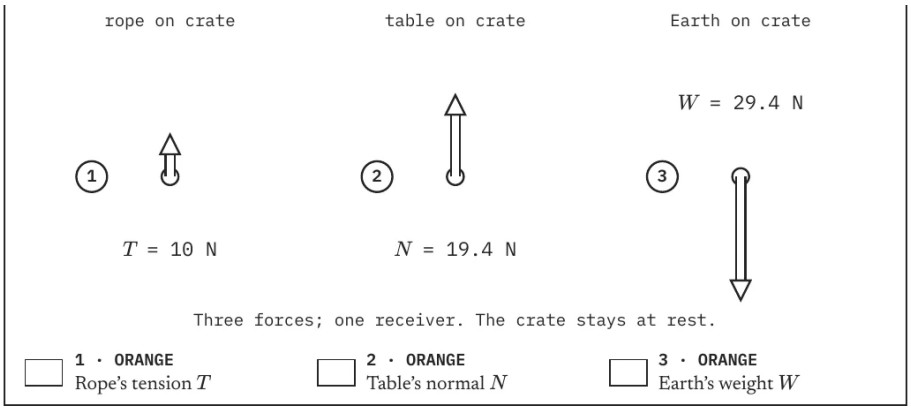
Also on your note page: Weight, Normal force, Tension.

B READ THE RELATIONSHIP

N + T − W = 0 for a crate at rest

As you read, find: source · receiver · contact.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 3 kg crate rests on a table. A vertical rope pulls up with 10 N . Find the table's normal force. Use g = 9.8 N/kg .

Given m = 3 kg T = 10 N up a y = 0

Rule W = mg . N + T − W = 0.

Work W = 3(9.8) = 29.4 N N = 29.4 − 10

Answer N = 19.4 N The rope shares the support.

E SAVE FOR THE FINAL MEETING

A forecast of XJ's changing motion needs real interactions, not a made-up motion force. Name the attracting body for each gravitational force before adding the forces.

Read the physics: Receiver

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Ask: Which object receives the force? What other object pushes or pulls it? A rope pulls by contact. Gravity pulls even without contact.

02 HOW IT WORKS

Mass is measured in kilograms. Weight is a force in newtons. On Earth, multiply mass by gravitational field strength to find weight. Earth pulls downward; a table pushes perpendicular to its surface.

03 WHAT TO CHECK

A rope may carry part of a crate's weight. If the crate stays at rest, all upward forces together balance the downward force.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 3 kg crate rests on a table. A vertical rope pulls up with 10 N . Find the table's normal force. Use g = 9.8 N/kg . N = 19.4 N The rope shares the support.

Model check: N + T − W = 0 for a crate at rest.

05 CONNECT TO XJ

A forecast of XJ's changing motion needs real interactions, not a made-up motion force. Name the attracting body for each gravitational force before adding the forces.

SAY IT BEFORE YOU WRITE

Who exerts the force, and on what object? The ___ exerts a force on the ___ toward ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Weight

ES: peso / PT: peso

Meaning in my words	How can you tell?
How would you explain weight to a friend? In my words, this means ___.	When a book rests on a table, which object supplies its weight? I can tell because ___.
Example or sketch	**Non-example and why**
Using 10 N/kg, what arrow represents the weight of a 3 kg book? My example shows ___.	A book slides down a ramp. Should its whole weight arrow point along the ramp? I would change ___ because ___.

Normal force

ES: fuerza normal / PT: força normal

Meaning in my words	How can you tell?
How would you explain normal force to a friend? In my words, this means ___.	A crate rests on a ramp. Which way does the ramp push the crate? I can tell because ___.
Example or sketch	**Non-example and why**
How would you draw the normal force on a book on a level table? My example shows ___.	Someone draws the normal force along the ramp. What would you rotate? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Tension

ES: tensión / PT: tração

Meaning in my words	How can you tell?
How would you explain tension to a friend? In my words, this means ___.	Which way does a slanted taut rope pull on a crate tied to its end? I can tell because ___.
Example or sketch	**Non-example and why**
Draw the force from a vertical cable on a hanging lamp. My example shows ___.	Could a loose rope push a cart forward when you move its free end toward the cart? I would change ___ because ___.

Receiver

ES: objeto que recibe la fuerza / PT: objeto que recebe a força

Meaning in my words	How can you tell?
How would you explain receiver to a friend? In my words, this means ___.	In “Earth pulls on the crate,” which object receives this force? I can tell because ___.
Example or sketch	**Non-example and why**
The rope pulls up on a lamp. On which object’s diagram do you draw that arrow? My example shows ___.	“The cart moves right.” Can you name a force receiver from that statement alone? I would change ___ because ___.

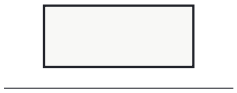
Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

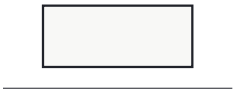
Draw a supported crate before and after an upward rope pull. Use the arrow lengths to show how the table's force changes.

Labels you can use: source · receiver · contact.

Upward support



Stronger rope pull



Tell a partner

The ___ exerts a force on the ___ toward ___.

Who is doing the pushing?

10

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Name the source and receiver of each force.

CHOOSE ONE JOB TOGETHER

SIMULATION ENGINEER ☐

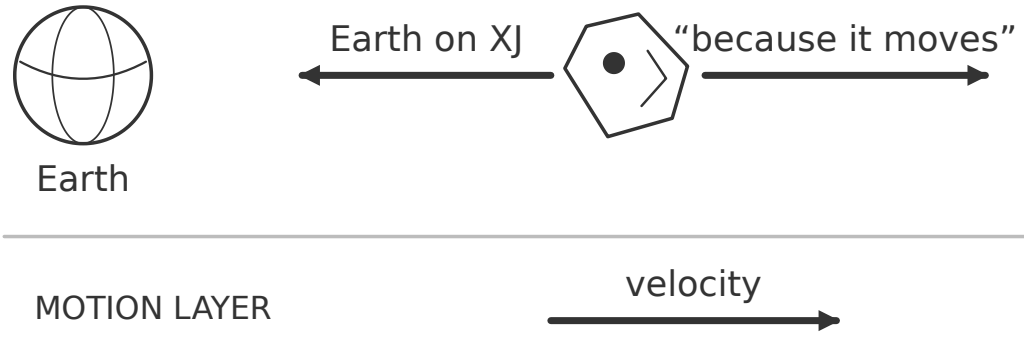
Approve the force layer for XJ. Name who exerts each force and what receives it. Keep motion separate.

SCIENCE GRAPHICS EDITOR ☐

Prepare the viewer graphic. Ask your partner who causes each proposed force before you approve its label.

Word help: source / receiver / gravity

An XJ animation shows gravity from Earth and a forward arrow labeled “because it moves.” The graphics desk needs approved force labels.



01 OUR FIRST CALL

Which force arrows should go live? Explain each arrow you keep or remove.

Keep ___ because ___ acts on ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Does the new information change the force layer? Explain your update.

Now we would say ___ because ___.

Switch speakers. Name another real source of force on XJ. Where would its arrow go, and what would you label it?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks